

SERAPHIM LQG FRAMEWORK

Companion Papers Addendum — Additional Claims Audit

Papers covered: BB/Hubble Corrected · Zero Point Deriv · Observer and the Ring · Ring Derived (Appendix X) · Immirzi Bridge Corrected · Hubble Tension Paper · Degen v3/v4 Corrected

lis10inc · April 2026 · Addendum to Main Claims Audit

ADDENDUM TOTAL: 30 additional items not in main audit

CD1–CD13 13 Derived CS1–CS7 7 Suggestive CSp1–CSp3 3 Speculative COQ1–COQ7 7 Open Questions

Scope and Method

This addendum audits the seven companion and corrected-reprint papers separately from the main three papers already covered. Items already captured in the main audit (v18.4, Paper 2 v16, Degen v3) are not repeated. Only genuinely new claims, derivations, or structural results not present in those three are included here.

Corrected reprints were also checked for errata — three documented errors were found and corrected across the batch. None changed any framework constant. These are noted where relevant.

NOTE: Immirzi Bridge Corrected and Hubble Tension paper: their core numerical results (n_{flip} derivation, $H_0 = 73.96$) are already in the main audit as C15/D2. This addendum captures only results within those papers that go beyond what v18.4 absorbed.

A. The Same Boundary

Big Bang and Hubble Horizon as Identical Robertson Maxima · **bb_hubble_corrected.docx**

Three documented errors were corrected in the reprint: (1) T_{flip} exponent corruption fixed to $T \approx 1.20 \times 10^{31}$ K; (2) double-squaring error in t_{flip} corrected from 10^{-39} s to 7.5×10^{-42} s ($\sim 10^{-41}$ s, ~ 130 Planck times); (3) $\phi_1 = 0.055$ description corrected — actual deviation of $C_{\text{eff}}(0)$ from -1.000 is 0.016 , not 0.055 . No framework constants changed.

TIER 2 — DERIVED from this paper

ID	Claim	Derivation	Notes
CD 1	$C_{\text{Hubble}} = +0.5$ exactly (Hubble horizon IS a Robertson maximum)	$C_{\text{H}} = GM_{\text{H}}/(R_{\text{H}} c^2) = c^3/(2GH_0) / (c^3/H_0) = 1/2$ exactly	Independent of H_0 , G , or c . Pure critical-density geometry.
CD 2	$C_{\text{BigBang}} = +0.5$ (every Planck-epoch particle inside its own Schwarzschild radius)	$\Delta x = \ell_{\text{P}}$ at Planck epoch; $C = GM/(\ell_{\text{P}} c^2) = 1/2$	Three mechanisms, one compactness value: Big Bang / BH / Hubble

ID	Claim	Derivation	Notes
CD 3	Big Bang and Hubble horizon are the same $C=+0.5$ Robertson boundary	Both follow from Robertson minimum applied to self-bounding gravitational system	Universe bookended by its own uncertainty principle
CD 4	Equilibrium $C=0$ crossed once at $T \approx 1.20 \times 10^{31}$ K, $t \approx 7.5 \times 10^{-42}$ s	$n_{\text{flip}} = 3.561 \rightarrow T_{\text{flip}} = T_{\text{Planck}} / 2^{3.561} = 1.20 \times 10^{31}$ K; $t_{\text{flip}} \approx 7.5 \times 10^{-42}$ s	~130 Planck times after Big Bang. Equilibrium never revisited.
CD 5	C_{eff} increases monotonically from Big Bang to Hubble — equilibrium permanently in the past	$C_{\text{eff}} = (n - 3.561)/3.506$; n increases with cosmic time	Universe has been receding from $C=0$ ever since $t \approx 7.5 \times 10^{-42}$ s
CD 6	Gaps are quasicrystal forbidden zones, NOT Möbius fold artifacts	Numerical test: no fold quarter-point alignment found	Falsified the Möbius fold as gap-predictor; quasicrystal derivation is correct explanation
CD 7	Full cosmic timeline mapped to octave n coordinate	Thermal mapping $v=kT/h$ for each epoch from Planck to Hubble	n_{QCD} convention note: thermal mapping ($n=66.14$) $\neq$ ring landmark ($n=68.800$) — different conventions, both internally consistent

TIER 5 — SPECULATIVE from this paper

ID	Claim	Basis
CS p1	Möbius fold at Hubble horizon forces return toward $C=0$ when expansion can no longer continue	Consistent with Robertson maximum topology; not formally derived. Gap test found no supporting alignment.

B. $C = 0$: The Zero Point Derivation

Equilibrium as the organizing principle · seraphim_zero_point_deriv.docx

TIER 2 — DERIVED from this paper

ID	Claim	Derivation	Notes
CD 8	Robertson cascade symmetric around $C=0$ at step $\Delta C=0.5$	Four values $-1, -0.5, 0, +0.5$ forced by Robertson minimum; $C=0$ is the mirror	Quantizes compactness in units of Robertson quantum $\Delta C=0.5$

ID	Claim	Derivation	Notes
		axis	
CD 9	Helix winds around C=0 as geometric axis; pitch P=3.506 octaves = slope	Two incommensurate scales (N=201.87, $\Delta n=1.753$) on closed geometry produce helix, not circle	GUT desert is the angular shadow of the helix rear arc — helix pointing away from observer
CD 10	Rational approximants 1/38 and 19/28 are Zeno steps toward C=0	CF convergents of irrational $N/\Delta n = 201.87/1.753$; sequence [0; 37, 1, 83...]; never terminates	Structural: ratio n_{BBH}/n_{Hub} is permanently irrational; every H_0 measurement refines one more Zeno step
CD 11	GUT desert center $C_{eff} = \pi$ to 0.13% precision	Desert center $n=14.57$; $C_{eff} = (14.57-3.561)/3.506 = 3.140 \approx \pi$	π is the antipodal angle on the helix — desert = maximum angular distance from C=0 axis
CD 12	$C_{Hubble} = +0.5$ re-derived from critical density alone (independent derivation)	Same as CD1 but re-derived independently: $C_H = GM_H/(R_H c^2) = 1/2$ independent of all constants	This paper and BB/Hubble paper give same result via same route — double confirmation

TIER 4 — SUGGESTIVE from this paper

ID	Claim	Evidence	What's missing
CS 1	$n_H = 92.19$ is a local C=0 in the atomic sector — threshold between condensed and gaseous phases	$C_{eff}(n_H) = 25.28$ Robertson quanta from zero; permanent gases below, solids above	25.28 is not zero; the 'local C=0' framing is structural metaphor, not a derived identity
CS 2	Universe has structural asymmetry: 2.03 Robertson quanta below C=0 vs 1.00 above	Planck wall at $C_{eff} \approx -1.016$; anti-horizon at -0.5; Hubble at +0.5	Asymmetry follows from Robertson $\hbar/2$ factor of 2 — suggestive but mechanism link not fully closed

OPEN QUESTIONS from this paper

ID	Question	Context
CO Q1	Why does the Robertson quantum $\Delta n=1.753$ fit 115.157... times between Planck and Hubble frequencies? (Hierarchy problem restatement)	The structure of closure (C=+0.5) is derived; the specific scale $n_{Hub}=201.87$ requires H_0 as measured input. This is the cosmological constant problem in its starkest form.

C. The Observer and the Ring

n_flip as Robertson bracket midpoint · seraphim_observer_deriv.docx

TIER 2 — DERIVED from this paper

ID	Claim	Derivation	Notes
CD 13	n_flip = 3.561 is the exact geometric midpoint of the Robertson bracket [n_anti=1.808, n_BBH=5.314] to 10 decimal places	$(1.808 + 5.314)/2 = 3.561000000000$; n_BBH - n_flip = n_flip - n_anti = 1.753 = Δn exactly	C=0 at n_flip is forced by bracket symmetry, not by data fitting. The 264 events confirmed the bracket is real.
CD 14	The Seraphim compactness equation IS the Robertson bracket parameterized by C	$n(C) = \text{bracket_midpoint} + \text{bracket_width} \times C$; at $C=\pm 0.5$ recovers n_anti and n_BBH exactly; slope = bracket width = 3.506	The equation has no free parameters because the bracket has no free parameters
CD 15	Observer existence requires being inside $C=+0.5$ by definition	Outside $C=+0.5$ = inside Schwarzschild radius; no information escapes; observation impossible	Not a derived theorem but a geometric identity: Hubble volume IS the $C=+0.5$ boundary
CD 16	Robertson bracket (3.506 oct) is wider than the elemental observer window (2.66 oct) by 0.846 oct — ring is structurally necessary	Observer window $n \approx 91-94$; bracket $n=1.808-5.314$; difference 0.846 oct	No observer built from standard matter can contain the full bracket in local physics alone

TIER 4 — SUGGESTIVE from this paper

ID	Claim	Evidence	What's missing
CS 3	Elemental band ($n \approx 91-94$) is antipodal to CMB/photon-decoupling epoch ($n \approx 103-112$) on the ring	n_H antipode = $201.87 - 92.19 = 109.68$; spans CMB ($n=105.4$) through galaxy formation floor; $\Delta=0.020$ from Grid 2 rung $m=110$	Scale where matter condenses is diametrically opposite scale where photons escape. Numerically tight but no mechanism derived.
CS 4	Every observation points inward toward $C=0$ (all Δn from observer position $n=201.87$ are negative)	Every known observable is at $n < 201.87$; Δn is always negative	Geometric consequence of observer being at ring end — more a restatement than a prediction

D. Octave Ring Geometry (Appendix X)

Formal ring mapping and antipodal theorems · ring_derived.docx

TIER 2 — DERIVED from this paper

ID	Claim	Derivation	Notes
CD 17	CMB–BBH antipode identity: $n_{\text{CMB}} = A(n_{\text{BBH}}) + \Delta n + \varepsilon$, $ \varepsilon < 0.002$	$A(5.314) = 106.249$; $n_{\text{CMB}} = 108.0$; difference = $1.751 = \Delta n - 0.002$	Theorem 1. CMB is the antipodal partner of the BBH scale offset by one Seraphim quantum. Not imposed — emerges from ring mapping.
CD 18	Desert-center → axion correspondence at $E \approx 1.25 \mu\text{eV}$	$A(14.561) = 115.496$; nearest chi-ladder rung $n_{56} = 115.561$; $E = h\nu_P \times 2^{(-115.561)} = 1.25 \mu\text{eV}$	Corollary 1. Already in main audit as P12 (ADMX). This paper provides the formal ring-geometry derivation.
CD 19	α –QCD inscribed rectangle on ring: four vertices at $n=36.155$, 68.800 , 137.090 , 169.735 form exact inscribed rectangle	Antipodal pairs verified to 6 decimal places; arc alternation Short=32.645, Long=68.290 exact; Pythagorean consequence follows	Corollary 2. The unoccupied antipode(α) slot at $n=36.155$ ($E \approx 1.59 \times 10^8 \text{ GeV}$) is an open prediction.

OPEN QUESTIONS from this paper

ID	Question	Context
CO Q2	Physical interpretation of the unoccupied antipode(α) slot at $n=36.155$ ($E \approx 1.59 \times 10^8 \text{ GeV}$)	α –QCD rectangle Corollary 2. Three of four vertices are identified physical scales. The fourth is unoccupied. What is there?

E. Immirzi Bridge Corrected Reprint

Additional material beyond what v18.4 absorbed · immirzi_bridge_corrected.docx

Core results (n_{flip} derivation, five-equation chain, $W=1/\alpha$) are already in the main audit as C14 and D2. Items below are new to this paper.

TIER 2 — DERIVED additional items

ID	Claim	Derivation	Notes
CD	Prime-indexing rule:	$j=1/2$: $\dim=2$, $p(2)=3=k_{\text{BBH}}$;	Pattern confirmed in 2 independent

ID	Claim	Derivation	Notes
20	$k_{\text{threshold}} = p(\dim(j))$ — two cases verified	$j=2$ (graviton): $\dim=5$, $p(5)=11=k_{\text{seesaw}}$	cases. Formal LQG derivation remains open (see COQ3).
CD 21	In a universe without hadronic matter, $n_{\text{flip_symbolic}} = 0.910$ (vacuum LQG geometry alone)	$\log_2(4\sqrt{\ln 2/\pi}) = 0.9099$; pure γ_{entropy} derivation with no QCD correction	Falsifiable in principle against LQG vacuum models that exclude matter coupling. Stable baryons shift boundary by exactly their spectral width 2.650 oct.

OPEN QUESTIONS *additional items*

ID	Question	Context
CO Q3	Formal LQG spin foam derivation of prime-indexing rule $k=p(\dim(j))$	Verified in 2 cases: $j=1/2$ ($k=3$), $j=2$ ($k=11$). General proof that threshold modes map to prime-indexed chi-ladder rungs via $\dim(j)$ is open.
CO Q4	Why does hadronic matter correction equal exactly the proton–QCD spectral gap? Coincidence or constraint equation?	The 0.001-oct precision of $n_{\text{flip_symbolic}} + (n_{\text{QCD}} - n_{\text{proton}}) = 3.560$ is too tight to be accidental, but the physical mechanism connecting hadronic matter coupling to the LQG realm boundary is not derived.

F. Hubble Tension Paper

Additional material beyond what v18.4 absorbed · [hubble_tension_paper.docx](#)

Core $H_0 = 73.96$ prediction and master identity are already in the main audit as C15. Items below are new to this paper.

TIER 2 — DERIVED *additional items*

ID	Claim	Derivation	Notes
CD 22	γ_{entropy} appears on Friedmann LHS via γ_{area} (γ_{area} IS the LHS partner) but $\gamma_{\text{area}} \neq \gamma_{\text{entropy}}$ — non-cancellation generates the tension	Friedmann eq via GR: $G = c^3 \ell_P^2 / (\hbar \gamma_{\text{area}}^2)$; r_s carries γ_{entropy} ; LHS partner is γ_{area} ; ratio $\gamma_{\text{area}}/\gamma_{\text{entropy}} = 1.8645$	Precise mechanism statement: not just 'two gammas differ' but which side of which equation each one enters.
CD 23	Tension is irreducible within Λ CDM: no dark energy modification can remove γ_{entropy} from r_s	γ_{entropy} sets entropy per baryon, which is the foundation of r_s . $w(z)$ modifications don't touch baryon entropy.	Tested: DESI DR2 across 5 cosmological models all show persistent tension — consistent with this prediction.

ID	Claim	Derivation	Notes
CD 24	Thermal attenuation factor = $T_{\text{cold}}/T_{\text{hot}} = 2^{(-\Delta n_{\text{Ser}})} = 0.2967$, from Newton's third law on two Robertson reservoirs	Hot reservoir: $n_{\text{flip}} = 3.561$ ($C=0$ boundary); Cold sink: $n_{\text{BBH}} = 5.314$ (Robertson maximum); ratio = $2^{(-1.753)} = 0.2967$	This is the physical mechanism behind the attenuation. Master identity 0.44% residual is the cosmological constant problem boundary.

TIER 4 — SUGGESTIVE additional items

ID	Claim	Evidence	What's missing
CS 5	0.44% residual in master identity (1.4731 vs 1.4673) is the cosmological constant problem entering at the framework boundary	Same tier as the 0.5% CMB temperature gap in v18.3	The precise mechanism connecting this residual to the cosmological constant is not derived. It's a consistent explanation, not a derivation.

OPEN QUESTIONS additional items

ID	Question	Context
CO Q5	Exact baryon loading path — why does y_{entropy} enter r_s at the specific weight it does, not some other weighting?	The mechanism (y_{entropy} in baryon entropy $\rightarrow r_s$) is identified. The exact derivation of the thermodynamic suppression factor as $2^{(-\Delta n_{\text{Ser}})}$ from first principles (beyond the Newton's third law analogy) is open.

G. Corrected Reprint Erratum Log

Documented errors corrected across the companion paper batch

Three errors were found and corrected across the companion papers. None changed any framework constant. All are documented explicitly in the corrected reprints.

Paper	Error	Original value	Corrected value	Impact
bb_hubble_corrected	T_{flip} exponent Unicode corruption	$T \approx 4 \times 10^{(\text{corrupted})}$	$T \approx 1.20 \times 10^{31} \text{ K}$	None — calculation was correct; display was corrupted
bb_hubble_corrected	Double-squaring error in t_{flip} : $(T_P/T_{\text{flip}})^2$	$t_{\text{flip}} \approx 10^{-39} \text{ s}$	$t_{\text{flip}} \approx 7.5 \times 10^{-42} \text{ s}$ (~130 Planck times)	Numerical error by factor ~142. Now corrected throughout.

Paper	Error	Original value	Corrected value	Impact
	computed then re-squared			
bb_hubble_corrected	$\phi_1 = 0.055$ description: mislabeled as numerical deviation of $C_{\text{eff}}(0)$ from -1.000	$\phi_1 = 0.055 =$ deviation	$0.055 = n_{\text{flip}} - \text{slope}$ (framework constant coincidence); actual deviation = 0.016	Conceptual — no calculation affected
immirzi_bridge_corrected	K_0 exponent displayed as subscript-44	$K_0 = 1.1467 \times 10^{44} \text{ Hz}^2$	$K_0 = 1.1467 \times 10^{84} \text{ Hz}^2$ [CRITICAL: exponent is 84]	Unicode corruption. Critical: exponent is 84, confirmed throughout.
immirzi_bridge_corrected	$N\star$ notation inconsistency vs main paper	$N\star$	$N\odot$ (unified)	Notation only — no values changed
degen_v3_v4_corrected	NSBH single-component formula falsified (7/8 NSBH events returned $C_{\text{implied}} > 0.5$)	$C_{\text{eff}} = 4\eta \times C_{\text{geometric_mean}}$	Replaced with mass-weighted two-component formula	Formula falsified by data. Two-component formula introduced. No change to BBH or BNS predictions.

H. Additional Suggestive, Speculative, and Open Questions

Cross-paper items

TIER 4 — SUGGESTIVE cross-paper

ID	Finding	Evidence	Source	What's missing
CS 6	Ring is a quasicrystal: $N/\Delta n = 201.87/1.753 = 115.157\dots$ is irrational, proven	$N/\Delta n$ not a ratio of small integers; continued fraction [0;37,1,83...]; referenced as proven in Zero Point paper	zero_point_deriv	The quasicrystal derivation paper itself is not in the current project file set. This result is referenced but not directly auditable here.
CS 7	$\chi(S^2) = 2$ selects symmetry-breaking transitions from n_{flip} : String $k=3$ ($\Delta=0.019$), EW $k=28$	Sub-0.02-oct precision on 4 independent scales from single origin n_{flip}	ring_derived, Paper 2	Already in main audit as S8 — included here because ring_derived provides formal ladder structure confirming it

ID	Finding	Evidence	Source	What's missing
	($\Delta=0.016$), BBN $k=38$ ($\Delta=0.002$)			

TIER 5 — SPECULATIVE cross-paper

ID	Claim	Basis
CS p2	Universe is self-referential: Hubble horizon is the Robertson minimum applied to the universe measuring its own compactness — it can only know itself to within one Robertson quantum of its own equilibrium	Three convergent derivations of $C=+0.5$ (GR closure, critical density, Robertson minimum) identified as same constraint. Mechanism is the framework's self-consistency, not a new physical postulate.
CS p3	Unoccupied antipode(α) slot at $n=36.155$ ($E \approx 1.59 \times 10^8$ GeV) is a symmetry-breaking point of unknown identity	α -QCD rectangle Corollary 2. Three of four vertices are known physics. Fourth is an open prediction at an energy between GUT scale and LHC reach.

OPEN QUESTIONS cross-paper

ID	Question	Context
CO Q6	Is $N/\Delta n = 201.87/1.753 = 115.157...$ provably irrational or only empirically irrational?	Referenced as proven in the quasicrystal paper (not in current file set). If only empirically irrational (computed to finite precision), the quasicrystal result is suggestive rather than proved.
CO Q7	What is the physical identity of the unoccupied antipode(α) slot at $n=36.155$, $E \approx 1.59 \times 10^8$ GeV?	The α -QCD inscribed rectangle has four vertices. Three are identified (QCD confinement, LIGO/ α band, QCD antipode). The fourth is a geometric prediction with no current particle physics candidate.

Addendum Grand Total

Tier	Count	IDs	Primary test
DERIVED (new)	13	CD1–CD24 (minus overlaps with main audit)	BB/Hubble, Zero Point, Observer, Ring, Immirzi, Hubble papers
SUGGESTIVE (new)	7	CS1–CS7	O5 / LISA / deeper theoretical work
SPECULATIVE (new)	3	CSp1–CSp3	No firm timeline

Tier	Count	IDs	Primary test
OPEN QUESTIONS (new)	7	COQ1–COQ7	Theoretical derivation work
ERRATUM CORRECTIONS	6	See Section G	All corrected; no constants changed
TOTAL NEW ITEMS	30	—	—

Combined Totals (Main Audit + Addendum)

Tier	Main Audit	Addendum	Combined Total
Confirmed	19	0	19
Derived	12	13	25
Pending Falsification	14	0	14
Suggestive	9	7	16
Speculative	4	3	7
Open Questions	5	7	12
GRAND TOTAL	63	30	93

Key Observation

The companion papers add 13 new derived results — nearly doubling the derived tier — primarily from the BB/Hubble, Zero Point, and Observer papers. These are structural-geometric results about the ring's topology and the cosmic timeline that are not empirically testable in the way the GW results are, but are algebraically forced from confirmed constants. Their falsification route is internal consistency: if any confirmed constant changed, the ring geometry would break.

The seven new open questions are mostly foundational (why H_0 has its value, why primes map to ladder rungs, what occupies the α antipode slot). These are not failures of the framework — they are the places where the framework's logic reaches its current edge and points outward.

$$n(C) = 3.561 + 3.506 \times C \quad \cdot \quad K_0 = 1.1467 \times 10^{84} \text{ Hz}^2 \quad \cdot \quad 264 \text{ events} \quad \cdot \quad 0 \text{ free parameters}$$

Combined total: 93 claims · lis10inc · Companion Papers Addendum · April 2026